

MATH 252 – MATLAB ASSIGNMENT 02

due by Friday, June 10

Instructions

The purpose of this assignment is to automate the numerical aspects of Slope Fields, Euler's Method, Phase line and Bifurcations. The goal is to edit the file `Labs02.FormatFile.mlx` while attempting the task at hand. After you are done, run your entire file, doublecheck your answers, and upload it to Canvas.

Exercises

1. **Slope Fields** Consider the differential equation given by

$$\frac{dy}{dt} = (t+2)(y+2)(y-3)^2.$$

- Define the function $f(t, y)$ that corresponds to the ordinary differential equation.
- Draw the slope field for $t \in [-1, 3]$ and $y \in [-4, 5]$ by using the adapted `quiver252` function on the supplementary code (bottom of the file).
- Use `ode45` built-in MATLAB function to obtain a numerical solution that solves for $y(0) = 1$. Plot the solution for $t \in [0, 3]$ in red.

2. **Euler's Method** Consider the initial value problem given by

$$\frac{dy}{dt} = e^t - t, \quad y(0) = 1.$$

The IVP has explicit solution given by

$$y(t) = e^t - \frac{t^2}{2}.$$

Additionally, in the Supplementary Code section of the file, which has a function named `EulerMethod` that outputs `t,y` given a certain input.

- Add the corresponding inputs necessary for Euler's method and implement the method inside the for loop.
- Define the corresponding $f(t, y)$ for the ODE

$$y' = f(t, y), \quad y(0) = 1$$

and the solution $Y(t)$ given above.

- Use Euler's Method and store results for each stepsize to estimate the solution of the IVP at the time $T_f = 2$.
 - $dt = 1$
 - $dt = 0.5$
 - $dt = 0.1$

Plot your results in different colors and using the option `'-o'`. This allows us to see precisely which points were computed. Write a legend of the graph that reflects the data displayed in the figure.

3. **Phase line** An autonomous differential equation is given by

$$x' = (x-1)^2(x+1).$$

- Find the constant solutions of $x' = f(x)$.
- In a single figure, plot for $t \in [0, 5]$, $y \in [-2, 2]$

- the constant solutions in blue,
- an unique solution in red in each distinct interval separated by constant solutions.

4. Bifurcations

4.1 A one-parameter ordinary differential equation is given by

$$y' = y^2 - (a + 2)y + 2a, \quad a \in [0, 5].$$

In the code, the parameter **a** can be controlled by a numerical slider. Each time **a** is changed, the code provided computes a sample of numerical solutions starting with $y(0) \in [-1, 6]$. There is a gradient of blue to green depending where it starts: blue corresponds to $y(0) = -1$, green to $y(0) = 6$.

- Determine how many equilibria solutions exist and their stability for $a = 0$.
- Determine how many equilibria solutions exist and their stability for $a = 2$.
- Determine how many equilibria solutions exist and their stability for $a = 5$.
- At which point does the differential equation undergoes a bifurcation, and which kind is it?

4.2 Repeat the analysis (code) for the differential equation

$$y' = (r - 2)y - y^2$$

using the numerical slider provided for **r**. At which point does the differential equation undergoes a bifurcation, and which kind is it?